

EXPERIMENT 7

Determination and plotting of Capacitance of isolated and concentric sphere with different inner and outer radius and dielectric constant of the dielectric mediums.

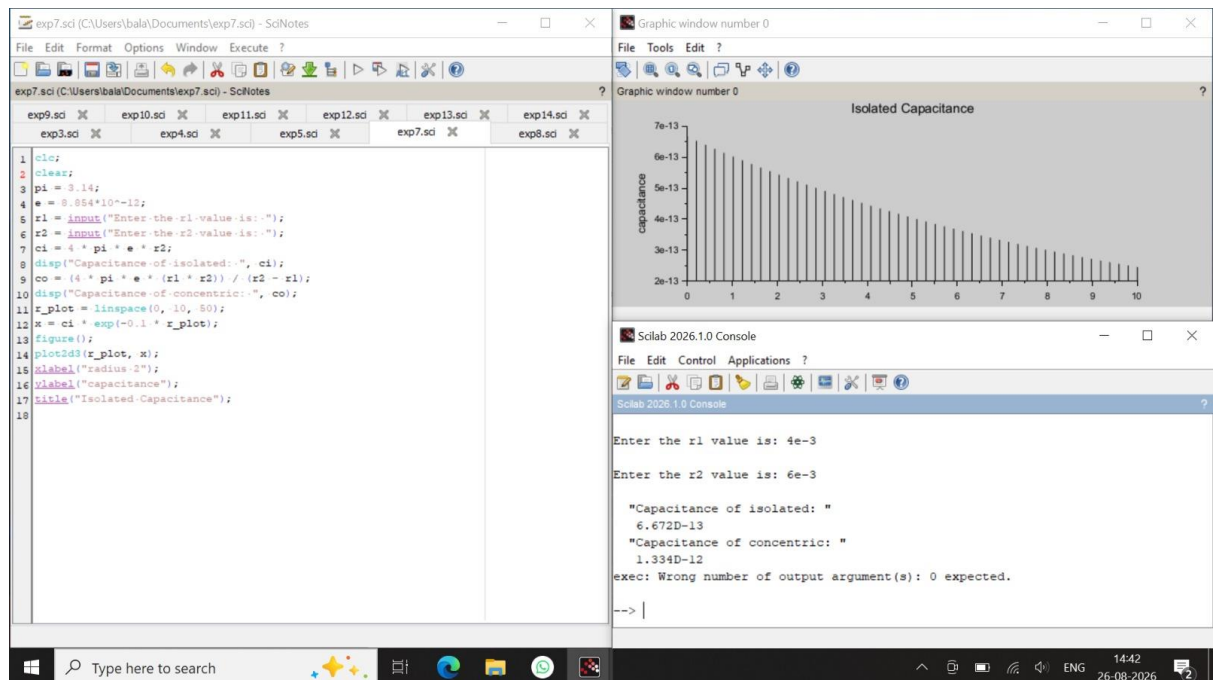
Aim:

To write program for Capacitance of an isolated sphere and two concentric sphere using SCILAB.

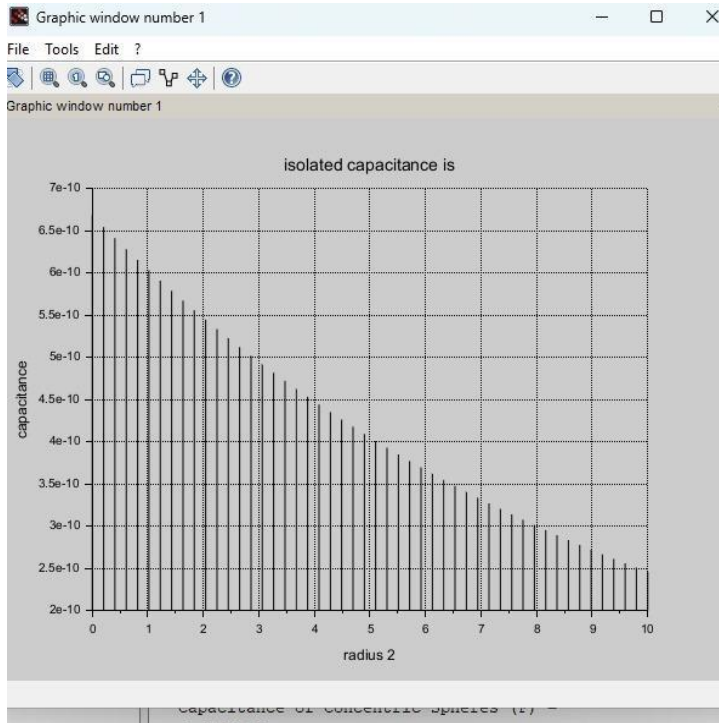
Software required:

- SCILAB version 6.0.1

Program:

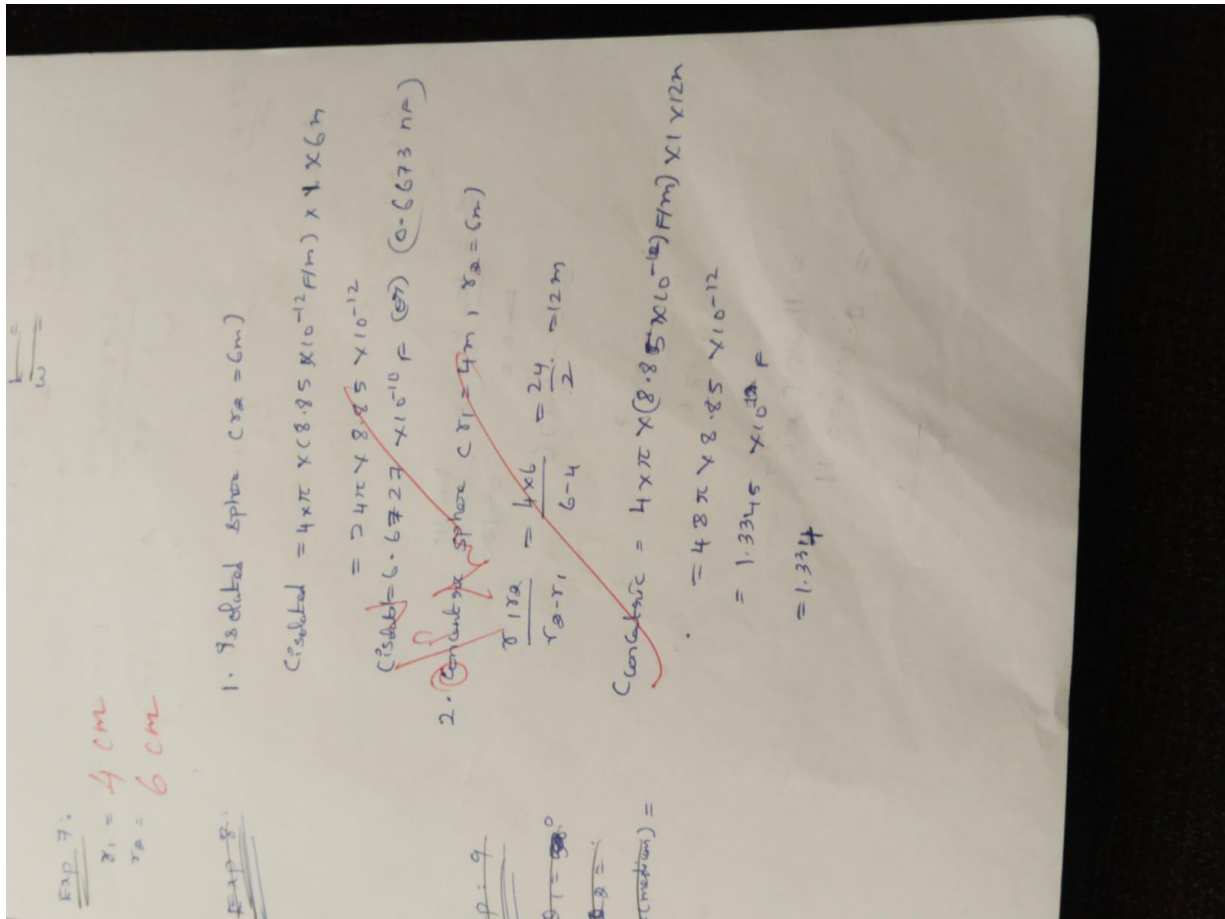


Output:



Practical output:

Theoretical calculation:



Enter the value of r1 (m): 3

Enter the value of r2 (m): 6

"Capacitance of Isolated Sphere (F) = "

6.676D-10

"Capacitance of Concentric Spheres (F) = "

6.676D-10

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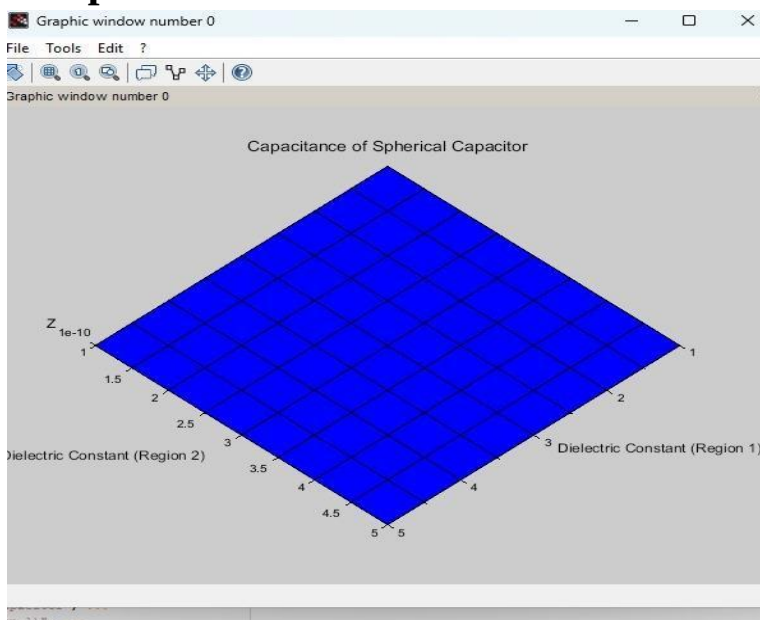
CASE 1:

-CONCENTRIC SPHERICAL CAPACITOR WITH DIFFERENT DIELECTRIC MEDIUMS

Program:

```
EX 7.sci (C:\Users\user\EX 7.sci) - SciNotes
File Edit Format Options Window Execute ?
EX 7.sci (C:\Users\user\EX 7.sci) - SciNotes
EX 7.sci
6 epsilon_0 = 8.854e-12;
7
8 // Radii (m)
9 R1 = 0.2;
10 Rm = 0.6;
11 R2 = 1.0;
12
13 // Relative permittivity values
14 epsilon_r1 = 1.0.5:5;
15 epsilon_r2 = 1.0.5:5;
16
17 // Create mesh
18 [X,Y] = meshgrid(epsilon_r1, epsilon_r2);
19
20 // Initialize capacitance matrix
21 C = zeros(X);
22
23 // Calculate capacitance
24 for i = 1:size(X,1)
25     for j = 1:size(X,2)
26         r1 = X(i,j);
27         r2 = Y(i,j);
28
29         C1 = 4 * pi * epsilon_0 * r1 * (R1 * Rm) / (Rm - R1);
30         C2 = 4 * pi * epsilon_0 * r2 * (Rm * R2) / (R2 - Rm);
31
32         // Equivalent capacitance
33         C(i,j) = (C1 * C2) / (C1 + C2);
34     end
35 end
36
37
38 // 3-D Surface Plot
39
40 plot3d(epsilon_r1, epsilon_r2, C);
41
42 title("Capacitance of Spherical Capacitor",...
43       "Dielectric Constant (Region-1)",...
44       "Dielectric Constant (Region-2)");
45
46 axis([0 5 0 5 0 1e-10]);
```

Output:



CASE 2:

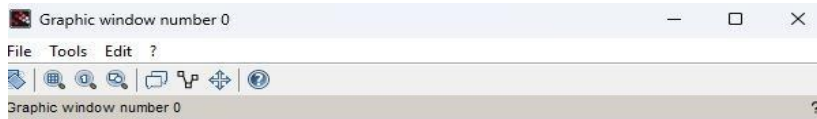
ISOLATED SPHERE

Program:

```
EX 7.sci (C:\Users\user\EX 7.sci) - SciNotes
File Edit Format Options Window Execute ?
EX 7.sci (C:\Users\user\EX 7.sci) - SciNotes
EX 7.sci X
2 clear;
3 clf;
4
5 //Radius from 0.01 m to 0.1 m
6 R = linspace(0.01, 0.1, 100);
7
8 //Relative permittivity values
9 epsilon = [1 2.5 3.5];
10
11 //Permittivity of free space
12 epsilon0 = 8.854e-12;
13
14 //Initialize capacitance matrix
15 C = zeros(3, length(R));
16
17 //Calculate capacitance
18 for i = 1:3
19     C(i,:) = 4 * pi * epsilon0 * epsilon(i) .* R;
20 end
21
22 //Plot graphs
23 plot(R, C(1,:));
24 plot(R, C(2,:));
25 plot(R, C(3,:));
26
27 //Change line colors
28 e = axes();
29 e.children(1).foreground = 4; //Blue
30 e.children(2).foreground = 2; //Green
31 e.children(3).foreground = 3; //Red
32
33 //Labels
34 xlabel("Radius- (m)");
35 ylabel("Capacitance- (F)");
36 title("Capacitance of Isolated Sphere vs Radius");
37
38 //Legend
39 legend("e=1 (Air)", "e=2.5", "e=3.5");
40
41 //Grid
42 xgrid();
```

Output:

CASE 3:



CONCENTRIC SPHERICAL CAPACITOR

Program:

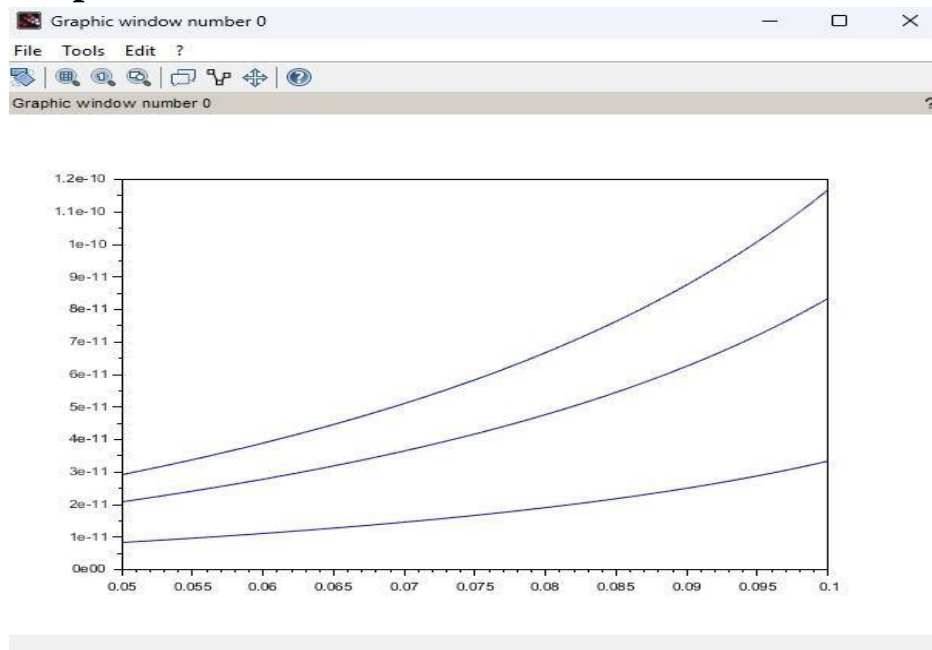
```

EX 7.sci (C:\Users\user\EX 7.sci) - SciNotes
File Edit Format Options Window Execute ?
EX 7.sci (C:\Users\user\EX 7.sci) - SciNotes
EX 7.sci
8 // Inner radius (m)
9 R1 = linspace(0.05, 0.10, 100);
10
11 // Outer radius (constant)
12 R2 = 0.15;
13
14 // Relative permittivity values
15 epsilon = [1 2.5 3.5];
16
17 // Initialize capacitance matrix
18 C = zeros(3, length(R1));
19
20 // Calculate capacitance
21 for i = 1:3
22     C(i,:) = (4 * pi * epsilon(i) * R1 * R2) ./ (R2 - R1);
23 end
24
25 // Plot first curve
26 plot(R1, C(1,:));
27
28 // Plot remaining curves
29 hold on;
30 plot(R1, C(2,:));
31 plot(R1, C(3,:));
32
33 // Set different colors
34 a = gca();
35 a.children(1).foreground = 'b'; // Blue
36 a.children(2).foreground = 'g'; // Green
37 a.children(3).foreground = 'r'; // Red
38
39 // Labels
40 xlabel('Inner Radius (m)');
41 ylabel('Capacitance (F)');
42 title('Capacitance of Concentric Spherical Capacitor vs Inner Radius');
43
44 // Legend
45 legend('epsilon=1', 'epsilon=2.5', 'epsilon=3.5');
46
47 grid on;
48

```

CASE 4:

Output:



Result:

Thus the program was verified for Capacitance of an isolated sphere and two concentric sphere using SCILAB was successfully.